

Compressing Files at the Shell Prompt

Compressed files use less disk space and download faster than large, uncompressed files. In Red Hat Linux you can compress files with the compression tools `gzip`, `bzip2`, or `zip`.

The `bzip2` compression tool is recommended because it provides the most compression and is found on most UNIX-like operating systems. The `gzip` compression tool can also be found on most UNIX-like operating systems. If you need to transfer files between Linux and other operating system such as MS Windows, you should use `zip` because it is more commonly used on those other operating systems.

Compression Tool	File Extension	Uncompression Tool
<code>gzip</code>	<code>.gz</code>	<code>gunzip</code>
<code>bzip2</code>	<code>.bz2</code>	<code>bunzip2</code>
<code>zip</code>	<code>.zip</code>	<code>unzip</code>

By convention, files compressed with `gzip` are given the extension `.gz`, files compressed with `bzip2` are given the extension `.bz2`, and files compressed with `zip` are given the extension `.zip`.

Files compressed with `gzip` are uncompressed with `gunzip`, files compressed with `bzip2` are uncompressed with `bunzip2`, and files compressed with `zip` are uncompressed with `unzip`.

Bzip2 and Bunzip2

To use `bzip2` to compress a file, type the following command at a shell prompt:

```
bzip2 filename
```

The file will be compressed and saved as `filename.bz2`.

To expand the compressed file, type the following command:

```
bunzip2 filename.bz2
```

The `filename.bz2` is deleted and replaced with `filename`.

You can use `bzip2` to compress multiple files and directories at the same time by listing them with a space between each one:

```
bzip2 filename.bz2 file1 file2 file3 /usr/work/school
```

The above command compresses `file1`, `file2`, `file3`, and the contents of the `/usr/work/school` directory (assuming this directory exists) and places them in a file named `filename.bz2`.



Tip

For more information, type `man bzip2` and `man bunzip2` at a shell prompt to read the man pages for `bzip2` and `bunzip2`.

Gzip and Gunzip

To use `gzip` to compress a file, type the following command at a shell prompt:

```
gzip filename
```

The file will be compressed and saved as `filename.gz`.

To expand the compressed file, type the following command:

```
gunzip filename.gz
```

The `filename.gz` is deleted and replaced with `filename`.

You can use `gzip` to compress multiple files and directories at the same time by listing them with a space between each one:

```
gzip -r filename.gz file1 file2 file3 /usr/work/school
```

The above command compresses `file1`, `file2`, `file3`, and the contents of the `/usr/work/school` directory (assuming this directory exists) and places them in a file named `filename.gz`.



Tip

For more information, type `man gzip` and `man gunzip` at a shell prompt to read the man pages for `gzip` and `gunzip`.

Zip and Unzip

To compress a file with `zip`, type the following command:

```
zip -r filename.zip filesdir
```

In this example, `filename.zip` represents the file you are creating and `filesdir` represents the directory you want to put in the new zip file. The `-r` option specifies that you want to include all files contained in the `filesdir` directory *recursively*.

To extract the contents of a `zip` file, type the following command:

```
unzip filename.zip
```

You can use `zip` to compress multiple files and directories at the same time by listing them with a space between each one:

```
zip -r filename.zip file1 file2 file3 /usr/work/school
```

The above command compresses `file1`, `file2`, `file3`, and the contents of the `/usr/work/school` directory (assuming this directory exists) and places them in a file named `filename.zip`.



Tip

For more information, type `man zip` and `man unzip` at a shell prompt to read the man pages for `zip` and `unzip`.

Archiving Files at the Shell Prompt

A `tar` file is a collection of several files and/or directories in one file. This is a good way to create backups and archives.

Some of the options used with the `tar` are:

- `-c` — create a new archive.
- `-f` — when used with the `-c` option, use the filename specified for the creation of the `tar` file; when used with the `-x` option, unarchive the specified file.
- `-t` — show the list of files in the `tar` file.
- `-v` — show the progress of the files being archived.
- `-x` — extract files from an archive.
- `-z` — compress the `tar` file with `gzip`.
- `-j` — compress the `tar` file with `bzip2`.

To create a `tar` file, type:

```
tar -cvf filename.tar files/directories
```

In this example, `filename.tar` represents the file you are creating and `files/directories` represents the files or directories you want to put in the archived file.

You can `tar` multiple files and directories at the same time by listing them with a space between each one:

```
tar -cvf filename.tar /home/mine/work /home/mine/school
```

The above command places all the files in the `work` and the `school` subdirectories of `/home/mine` in a new file called `filename.tar` in the current directory.

To list the contents of a `tar` file, type:

```
tar -tvf filename.tar
```

To extract the contents of a `tar` file, type:

```
tar -xvf filename.tar
```

This command does not remove the `tar` file, but it places copies of its contents in the current working directory.

Remember, the `tar` command does not compress the files by default. To create a tarred and bziped compressed file, use the `-j` option:

```
tar -cjvf filename.tbz
```

`tar` files compressed with `bzip2` are conventionally given the extension `.tbz`.

This command creates an archive file and then compresses it as the file `filename.tbz`. If you uncompress the `filename.tbz` file with the `bunzip2` command, the `filename.tbz` file is removed and replaced with `filename.tar`.

You can also expand and unarchive a `bzip tar` file in one command:

```
tar -xjvf filename.tbz
```

To create a tarred and gzipped compressed file, use the `-z` option:

```
tar -czvf filename.tgz
```

`tar` files compressed with `gzip` are conventionally given the extension `.tgz`.

This command creates the archive file `filename.tar` and then compresses it as the file `filename.tgz`. (The file `filename.tar` is not saved.) If you uncompress the `filename.tgz` file with the `gunzip` command, the `filename.tgz` file is removed and replaced with `filename.tar`.

You can expand a `gzip tar` file in one command:

```
tar -xzvf filename.tgz
```



Tip

Type the command `man tar` for more information about the `tar` command.